and choose three courses from the following list:

EENG509	SPARSE SIGNAL PROCESSING	3.0
EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	3.0
EENG511	CONVEX OPTIMIZATION AND ITS ENGINEERING APPLICATIONS	3.0
EENG514	DATA SCIENCE FOR ELECTRICAL ENGINEERING	3.0
EENG517	THEORY AND DESIGN OF ADVANCED CONTROL SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0
EENG589	DESIGN AND CONTROL OF WIND ENERGY SYSTEMS	3.0

Required Core: Power and Energy Systems Track

All students must choose four courses from the following list:

EENG570	ADVANCED HIGH POWER ELECTRONICS	3.0
EENG573	ELECTRIC POWER QUALITY	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	
EENG581	POWER SYSTEM OPERATION AND MANAGEMENT	3.0
EENG584	POWER SYSTEM RISK MANAGEMENT	3.0
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	
EENG587	POWER SYSTEMS PROTECTION AND RELAYING	3.0
EENG588	POWER SYSTEM ECONOMICS AND ELECTRICITY MARKETS	3.0

Professional Online Masters in Electrical Engineering

The professional master's degree is designed to train and target recent graduates or mid-career professionals with a B.S. in electrical engineering or a related field in physics or applied sciences. The program is composed of 3 stackable certificates plus a required graduate-level mathematics course (MEGN502). To complete the professional master's degree the student must complete 30 credits as outlined below.

Students may also register for the professional masters at the outset and take a mixture of courses (from the tracks) in any order. Should the student elect to register for the full masters, then certificates will not be awarded on completion of the full degree. For these students, we refer to the certificates as 'tracks' as no certificates will be awarded.

Certificate 1 / Track 1: Information and System Sciences – students must complete 9 hours of coursework as follows:

EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0

Certificate 2 / Track 2 : Microwave Engineering – students must complete 9 hours of coursework as follows:

EENG530	PASSIVE RF & MICROWAVE DEVICES
EENG532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING
EENG533	ACTIVE RF & MICROWAVE DEVICES

Certificate 3 / Track 3: Power & Energy Systems – students must complete 9 hours of coursework from among the following courses:

EENG570	ADVANCED HIGH POWER ELECTRONICS	3.0
EENG577	ADVANCED ELECTRICAL MACHINE DYNAMICS FOR SMART-GRID SYSTEMS	
EENG585	AI FOR POWER AND RENEWABLE ENERGY SYSTEMS	3.0
EENG588	POWER SYSTEM ECONOMICS AND ELECTRICITY MARKETS	3.0

Graduate certificate in information and system sciences

The post-baccalaureate certificate program in Information and System Sciences is targeted to train recent graduates or mid-career professionals with a basic knowledge of modeling and extracting information from signals, systems, or data sets.

To earn the post-baccalaureate certificate program in Information and System Sciences, students must complete 9 hours of coursework as follows:

Required Courses:

EENG510	ADVANCED DIGITAL SIGNAL PROCESSING	3.0
EENG515	MATHEMATICAL METHODS FOR SIGNALS AND SYSTEMS	3.0
EENG519	ESTIMATION THEORY AND KALMAN FILTERING	3.0

While all courses are to be delivered online, students that are interested in attending an on-campus class may substitute one of the above courses with one of the Information and Systems Sciences track electives as listed in our Graduate Catalog.

Graduate certificate in microwave engineering

The online certificate program in Microwave Engineering is targeted to train recent graduates or mid-career professionals with a B.S. in electrical engineering or a related field in physics or applied sciences with a basic knowledge of electromagnetic theory, specifically for handling the challenges and demands of modern microwave systems.

To complete the online certificate program in Microwave Engineering, students must complete 9 hours of coursework as follows:

Required Courses:

EENG530	PASSIVE RF & MICROWAVE DEVICES
EENG532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING
EENG533	ACTIVE RF & MICROWAVE DEVICES